The Optimization Problem of the Solar Tracking (Heliostat) Mirror Field

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Translator's note: *This is a faithful translation. Equation numbering, symbols, and table values follow the source as printed. A small number of evident typographical inconsistencies in the original are flagged in notes like this one but were NOT changed in the reproduced tables.*

Abstract

Addressing the problem of optimizing heliostat positions in a heliostat field, this paper analyzes and optimizes the spatial distribution of the heliostats so as to capture as much solar energy as possible, based on an optimization model together with a heliostat-optimization model. The heliostats across the field are handled with a layered, block-wise (stratification–segmentation) approach; MATLAB is used to build a polar-coordinate model of the heliostat positions, and a position-allocation model is established by modeling the angular and positional deviations between each heliostat and the absorption tower. The method copes effectively with the complexity introduced by adjusting a large number of heliostats and exhibits greater simplicity and accuracy.

Keywords: Heliostat Field, MATLAB, Optimization Model, Heliostat Position Planning

1. Introduction

Since the Industrial Revolution, humankind's intensive use of fossil fuels such as coal, oil, and natural gas has driven rapid industrialization, while over-exploitation and a near-frantic scramble for these precious resources have produced a global energy crisis [1]. As industrialization advanced, daily life became progressively mechanized, giving rise to automobiles and other modern industrial products. Consumption data indicate that, as of 2023, global resource trade amounted to roughly 31 trillion USD, of which petroleum trade accounted for about 17.55% [2].

However, the large-scale exploitation and inefficient use of fossil fuels have had serious consequences: they have intensified global warming and threatened human survival, making the development of new energy urgent. To safeguard ecological, social, and economic sustainability, countries began actively developing renewable energy as early as the start of the twentieth century [3]. As an inexhaustible renewable resource, solar energy has rapidly become a focus of research and attention [4].

With its powerful radiant energy and renewability, solar energy occupies an important place in new-energy development. As China has advanced its goals for green, sustainable development, vigorously developing solar and other renewable energy has become an important national topic [5], and solar thermal (concentrating solar power, CSP) generation has met unprecedented opportunities.

Technically, although photovoltaic generation has encountered bottlenecks that slow improvements in radiation-utilization efficiency, sustained research at home and abroad—together with advances in solar-concentration technology and in the heat-absorbing capacity of materials—has markedly improved the energy-conversion efficiency of CSP, whose theoretical efficiency now far exceeds that of PV [6][7].

In policy terms, as China's economy entered the "new normal" and faced widespread curtailment of coal, PV, and wind power, the state introduced benchmark feed-in tariffs for CSP and approved several demonstration CSP projects under the "Twelfth Five-Year Plan." This reflects the government's determination to develop CSP and signals CSP as an important link in the country's energy transition [8].

China's vast territory harbors abundant solar resources, so building tower CSP stations can effectively harvest them. A tower CSP system works mainly by reflecting and concentrating sunlight, via an efficient and precise heliostat field, onto a receiver at the top of the tower; the heat is stored and then converted by a heat-transfer device into mechanical energy that drives a turbine generator [9]. As a key component of the station, the reasonable distribution of the heliostats is very important, and this paper studies that distribution-optimization problem in depth (the tower CSP system is shown in Figure 1).

[Figure 1. Tower solar thermal power generation — photograph in the original.]

2. Model Establishment and Solution

2.1. Establishment of Model 1

(1) Average cosine efficiency

From the appendix, the cosine efficiency η_{cos} of a single heliostat is given by Eq. (1):

$$\eta_{cos} = 1 - \eta_{(cosine\ loss)} \quad (1)$$

where $\eta_{(cosine\ loss)}$ is the cosine-loss rate. Cosine loss is the energy reduction caused by the non-parallelism between the direction of the incident sunlight and the normal of the heliostat's light-collecting aperture. Letting θ be the angle between the incident sunlight and the aperture normal, by Eqs. (2)–(3) the cosine loss can be written as:

$$\eta_{(cosine\ loss)} = \cos\vartheta \quad (2)$$

$$\vartheta = (\alpha_s - \arcsin(h / d_{HR})) / 2 \quad (3)$$

where α_s is the solar altitude angle, h is the vertical height between the heliostat center and the collector center, and d_{HR} is the straight-line distance between the heliostat center and the collector center. Using α_s and d_{HR} as variables, MATLAB substitutes numerical values to compute, for each ring layer, the total sunlight received; summing over all layers and averaging yields the field's average cosine efficiency over a day.

Translator's note: *Equations (1)–(2) as printed give $\eta_{cos} = 1 - \cos\vartheta$. Physically the cosine efficiency should equal $\cos\vartheta$; the reported table values are consistent with $\cos\vartheta$, so the printed definition appears to be a typographical inversion.*

(2) Average shadow-and-blocking efficiency

Shadow-and-blocking loss has three parts [10]: ① shadow loss cast by the tower onto the field; ② "shadow loss," where sunlight bound for a rear heliostat is blocked by the heliostat in front; ③ "blocking loss," where light reflected by a rear heliostat is intercepted by a front heliostat and fails to reach the receiver. A heliostat's shadow may fall on another mirror, or reflected light may strike the back of another heliostat, producing loss; a shaded surface receives much less energy and converts little to output. The loss is computed by Eq. (4) and the efficiency by Eq. (5):

$$\left(L - h_{\text{(center distance)}} \right) / L = \eta_{\text{(blocking loss)}} / b \quad (4)$$

$$\eta_{sb} = 1 - \eta_{\text{(blocking loss)}} \quad (5)$$

where L is the shadow length cast when the mirror is illuminated, $h_{\text{(center distance)}}$ is the horizontal distance between the centers of the front and rear mirrors, $\eta_{\text{(blocking loss)}}$ is the shadow-blocking loss, b is the plate (mirror) length, and η_{sb} is the shadow-blocking efficiency. Using MATLAB on the appendix data, the horizontal center-to-center distances between adjacent layers and the shadow lengths under illumination are computed to obtain each mirror's shadow-blocking efficiency; summing and averaging gives the field average.

(3) Average truncation efficiency

Truncation efficiency is the percentage of the field's concentrated energy that the receiver actually captures, given by Eq. (6):

$$\eta_{trunc} = Q_{collector} / \left(Q_{\text{(total reflected)}} - Q_{\text{(shadow/blocking)}} \right) \quad (6)$$

Because the receiver cannot capture all incident energy, its captured energy is not the whole. After analyzing this optical phenomenon, adjusting the incidence angle α yields each layer's truncation efficiency. The receiver-captured energy is given by Eq. (7) and the total mirror-reflected energy by Eq. (8):

$$Q_{collector} = Q_{\text{(total reflected)}} \times \eta_{ref} \quad (7)$$

$$Q_{\text{(total reflected)}} = DNI \times (1 - \cos\alpha) \times S \quad (8)$$

where η_{ref} is the mirror reflectivity ($\eta_{ref} = 0.92$), S is the heliostat size ($S = 6\times6\ m^2$), and DNI is the direct normal irradiance. Substituting each layer's incidence angle gives the truncation efficiency of all heliostats; summing and averaging gives the field average.

(4) Average optical efficiency

By the appendix formula, the optical efficiency is given by Eq. (9):

$$\eta = \eta_{sb} \cdot \eta_{cos} \cdot \eta_{at} \cdot \eta_{trunc} \cdot \eta_{ref} \quad (9)$$

From (1)–(3) the shadow-blocking, cosine, and truncation efficiencies of each heliostat are known; with $\eta_{ref} = 0.92$, only the atmospheric transmittance η_{at} remains, which depends on d_{HR} via Eq. (10):

$$\eta_{at} = 0.99321 - 0.0001176 \cdot d_{HR} + 1.97\times10^{-8} \cdot d_{HR}^2 \quad (d_{HR} \le 1000) \quad (10)$$

Since the install height is 4 m and all coordinates are given, d_{HR} (hence η_{at}) follows from the Pythagorean theorem and trigonometry. Computing each heliostat's optical efficiency, summing, and dividing by the total number gives the field average optical efficiency.

(5) Average output power per unit mirror area

The field output thermal power E_field is the product of DNI, each mirror's area, and its optical efficiency, by Eq. (11):

$$E_{field} = DNI \cdot \sum_{i=1}^{N} A_i \cdot \eta_i \quad (11)$$

where A_i is the i-th heliostat's light-collecting area and η_i its optical efficiency. Since each heliostat is 6×6 m ($A_i = 36$), Eq. (12):

$$E_{field} = 36 \cdot DNI \cdot \sum_{i=1}^{N} \eta_i \quad (12)$$

Dividing the thermal output power by the total number of heliostats gives the per-unit-area average output power, Eq. (13):

$$\bar{E}_{field} = E_{field} / N \quad (13)$$

where N is the total number of heliostats. Per the problem requirements, the model is solved and the results are reported in Tables 1 and 2.

Table 1. Average optical efficiency and output power of Model 1 on the 21st of each month.

Date	Avg optical eff.	Avg cosine eff.	Avg shadow-blocking eff.	Avg truncation eff.	Per-unit-area avg output (kW/m²)
Jan 21	0.5825	0.9255	0.821	0.8392	776.751
Feb 21	0.5849	0.9520	0.833	0.8388	776.35
Mar 21	0.5832	0.9756	0.835	0.8482	776.983
Apr 21	0.5899	0.9921	0.837	0.8463	776.71
May 21	0.5814	0.9978	0.832	0.8449	776.533
Jun 21	0.5867	0.9986	0.83	0.8436	776.859
Jul 21	0.5866	0.9978	0.834	0.8427	776.374
Aug 21	0.5827	0.9916	0.829	0.8418	776.23
Sep 21	0.5884	0.9745	0.827	0.8411	776.20
Oct 21	0.5839	0.9488	0.823	0.8405	776.19
Nov 21	0.5873	0.9231	0.82	0.84	775.958
Dec 21	0.5891	0.9132	0.818	0.8396	775.725

Table 2. Annual average optical efficiency and output power of Model 1.

Annual avg optical eff.	Annual avg cosine eff.	Annual avg shadow-blocking eff.	Annual avg truncation eff.	Annual avg output (MW)	Per-unit-area annual avg output (kW/m²)
0.5856	0.9659	0.8282	0.8422	48.7945	776.4052

Translator's note: *The per-unit-area values in the source (e.g. 776.4052) are labeled kW/m² but are numerically in W/m² (≈ 0.776 kW/m²). Reproduced here exactly as printed.*

2.2. Establishment of Model 2

Suppose the average output thermal power is raised to 60 MW. By the control-variable method and Eq. (12), the main factors affecting E_field are the total mirror area and the collection efficiency. Repeated trials show that changing install height and number alters the collection efficiency little; therefore changing the mirror area brings the output close to the rated value, and the heliostat size at rated power can be determined.

Considering both the change in output power and the change in total area, extensive MATLAB tuning finds that with a mirror size of 6×7 m², install height 4 m, and the number adjusted to 1700 — uniformly distributed on 18 concentric circles outside the 100 m tower-exclusion zone — the field's annual average output is very close to the rated power, so the per-unit-area annual average output is maximized. Applying the Model-1 procedure with these settings yields the remaining Problem-2 parameters (Tables 3, 4, and 5).

Table 3. Average optical efficiency and output power of Model 2 on the 21st of each month.

Date	Avg optical eff.	Avg cosine eff.	Avg shadow-blocking eff.	Avg truncation eff.	Per-unit-area avg output (kW/m²)
Jan 21	0.5830	0.9251	0.825	0.8396	776.662
Feb 21	0.5845	0.9532	0.835	0.8324	776.521
Mar 21	0.5866	0.9748	0.839	0.8442	776.328
Apr 21	0.5841	0.9915	0.834	0.8476	775.291
May 21	0.5895	0.9985	0.831	0.8405	776.852
Jun 21	0.5823	0.9935	0.836	0.8442	776.395
Jul 21	0.5847	0.9947	0.823	0.8491	775.269
Aug 21	0.5811	0.9975	0.824	0.8455	775.841
Sep 21	0.5864	0.9742	0.821	0.8436	776.208
Oct 21	0.5865	0.9488	0.834	0.8501	776.526
Nov 21	0.5842	0.9235	0.819	0.8305	777.129
Dec 21	0.5894	0.9234	0.814	0.8426	776.891

Table 4. Annual average optical efficiency and output power of Model 2.

Annual avg optical eff.	Annual avg cosine eff.	Annual avg shadow-blocking eff.	Annual avg truncation eff.	Annual avg output (MW)	Per-unit-area annual avg output (kW/m²)
0.5852	0.9666	0.8279	0.8425	60	776.3261

Table 5. Design-parameter table for Model 2.

Absorption-tower position	Heliostat size (W × H)	Install height (m)	Total number	Total mirror area (m²)
(0, 0)	6 × 7	4	1700	32

Translator's note: *Table 5 lists the total mirror area as "32 m²," which is evidently a typo; 1700 mirrors of 6×7 = 42 m² give ≈ 71,400 m². Reproduced as printed.*

2.3. Establishment of Model 3

Allowing heliostat sizes and install heights to differ, the rated power is still 60 MW; this part designs the parameters to maximize the average output. A "stratified-gradient" strategy is used: each layer's install height increases progressively. Install height is limited to 2–6 m and must keep the mirror from touching the ground while rotating about the horizontal axis — a constraint that is fully respected.

After repeated validation, the heliostat number and arrangement are kept identical to Problem 2 (where the output is most reasonable and maximal). The 18 circles outside the 100 m zone are divided into three zones: (1) circles 1–6; (2) circles 7–12; (3) circles 13–18. Within each zone all circles share the same size and install height. The three zones' average output powers are summed to give the field total, Eq. (14):

$$\bar{E}\_output = \Sigma(i=1..3)\ E\_i \qquad (14)$$

Continued tuning determines the specifications and install heights that maximize the per-unit-area annual average output at 60 MW; the three parts' specifications are shown in Table 6.

Table 6. Sizes and install heights of the heliostats in the three parts of Model 3.

Part	Size	Install height
Part 1	6 m × 7 m	4 m
Part 2	7 m × 7 m	5 m
Part 3	7 m × 7 m	6 m

Part 1's chosen specification equals Model 2's, so Part 1's data equal Tables 3 and 4. Part 2's parameters appear in Tables 7, 8, and 9.

Table 7. Average optical efficiency and output power on the 21st of each month — Part 2 of Model 3.

Date	Avg optical eff.	Avg cosine eff.	Avg shadow-blocking eff.	Avg truncation eff.	Per-unit-area avg output (kW/m²)
Jan 21	0.5267	0.9364	0.77	0.8609	702.3779
Feb 21	0.5253	0.9331	0.723	0.8685	702.852
Mar 21	0.5239	0.9336	0.724	0.8635	702.395
Apr 21	0.5275	0.9323	0.721	0.8647	702.269
May 21	0.5211	0.9324	0.734	0.8642	702.841
Jun 21	0.5247	0.9321	0.719	0.8633	702.836
Jul 21	0.5299	0.9334	0.764	0.8645	702.539
Aug 21	0.5238	0.9319	0.778	0.8678	702.334
Sep 21	0.5271	0.9323	0.779	0.8626	702.911
Oct 21	0.5247	0.9324	0.773	0.8693	702.339
Nov 21	0.5217	0.9321	0.777	0.8597	702.886
Dec 21	0.5217	0.9334	0.769	0.862	702.634

Table 8. Annual average optical efficiency and output power — Part 2 of Model 3.

Annual avg optical eff.	Annual avg cosine eff.	Annual avg shadow-blocking eff.	Annual avg truncation eff.	Annual avg output (MW)	Per-unit-area annual avg output (kW/m²)
0.5248	0.9329	0.7526	0.8642	61	702.601

Table 9. Design-parameter table — Part 2 of Model 3.

Absorption-tower position	Heliostat size (W × H)	Install height (m)	Total number	Total mirror area (m²)
(0, 0)	7 × 7	5	594	49

Part 3’s parameters appear in Tables 10, 11, and 12.

Translator's note: *As with Table 5, the “total area” entries in Tables 9 and 12 (“49”) are evidently per-mirror areas, not field totals (594 × 49 ≈ 29,106 m²; 728 × 49 ≈ 35,672 m²). The running text also states Part 3’s height as 4 m, while Tables 6 and 12 give 6 m. Reproduced as printed.*

Table 10. Average optical efficiency and output power on the 21st of each month — Part 3 of Model 3.

Date	Avg optical eff.	Avg cosine eff.	Avg shadow-blocking eff.	Avg truncation eff.	Per-unit-area avg output (kW/m²)
Jan 21	0.5271	0.9372	0.777	0.8607	703.245
Feb 21	0.5238	0.9325	0.773	0.8632	703.662
Mar 21	0.5277	0.9335	0.775	0.8648	703.521
Apr 21	0.5294	0.9339	0.777	0.8615	703.328
May 21	0.5217	0.9334	0.772	0.8685	703.291
Jun 21	0.5217	0.9331	0.763	0.8635	703.852
Jul 21	0.5269	0.9336	0.764	0.8647	703.395
Aug 21	0.5261	0.9323	0.769	0.8675	703.269
Sep 21	0.5295	0.9324	0.777	0.8642	703.841
Oct 21	0.5247	0.9321	0.773	0.8688	703.208
Nov 21	0.5285	0.9334	0.762	0.8635	703.526
Dec 21	0.5235	0.9319	0.778	0.8634	703.129

Table 11. Annual average optical efficiency and output power — Part 3 of Model 3.

Annual avg optical eff.	Annual avg cosine eff.	Annual avg shadow-blocking eff.	Annual avg truncation eff.	Annual avg output (MW)	Per-unit-area annual avg output (kW/m²)
0.5259	0.9333	0.7717	0.8645	62	703.439

Table 12. Design-parameter table — Part 3 of Model 3.

Absorption-tower position	Heliostat size (W × H)	Install height (m)	Total number	Total mirror area (m²)
(0, 0)	7 × 7	6	728	49

2.4. Method Comparison

Designing a heliostat field is complex, spanning optics, structural mechanics, materials science, and control engineering. Existing optimization methods mainly include:

1) Optical-performance optimization. Ray tracing simulates the sun’s rays at different times of day and seasons onto the heliostats to optimize the layout for maximum collection efficiency; mirror material and process optimization selects high-reflectivity, durable materials and advanced manufacturing; mirror-cleanliness maintenance cleans mirrors regularly to preserve high reflectivity.

2) System control-precision optimization. Tracking-system improvement uses more precise sensors and controls (e.g., dual-axis tracking) so the mirror always points accurately at the sun; software/algorithm optimization develops efficient algorithms to predict the sun’s position and adjust mirror angles in real time for maximum concentration.

Of these, the most important is predicting the sun's position via efficient algorithms that model its daily and seasonal path so the mirrors continually adjust for maximum capture. This paper instead uses a multi-size optimization, configuring heliostats of different sizes to suit different geographic and climatic conditions, and performs an overall layout optimization considering inter-heliostat distance and angle. The key difference is that the present approach needs neither frequent angle adjustment nor complex sun-position prediction; it only adjusts the sizes, heights, and inter-zone distances and angles. By assigning different zones different sizes, heights, and angles, mutual shadowing between zones during the sun's daily course is avoided, each zone attains its best illumination duration, and each zone's capture rate is maximized.

3. Conclusion

This study designs three parts of the heliostat field, assigning parameters to each. Combining Models 1, 2, and 3 shows that, with differing sizes and install heights, dividing the field into three parts and designing each part's parameters maximizes the per-unit-area annual average output thermal power at the rated power. The final integrated data appear in Tables 13, 14, and 15; Table 6 details the sizes used. In sum, through model analysis and data integration, the paper reaches conclusions on how to design the field layout to maximize heat output when heliostat sizes and install angles vary, offering a useful reference for heliostat-field design.

Table 13. Average optical efficiency and output power on the 21st of each month — integrated result, Question 3.

Date	Avg optical eff.	Avg cosine eff.	Avg shadow-blocking eff.	Avg truncation eff.	Per-unit-area avg output (kW/m²)
Jan 21	0.5456	0.9329	0.790	0.8537	727.4283
Feb 21	0.5445	0.9396	0.777	0.8547	727.6783
Mar 21	0.5461	0.9473	0.779	0.8567	727.4147
Apr 21	0.5470	0.9526	0.777	0.8679	726.9627
May 21	0.5441	0.9548	0.779	0.8577	727.6613
Jun 21	0.5429	0.9529	0.7726	0.857	727.6943
Jul 21	0.5472	0.9539	0.7837	0.8594	727.0677
Aug 21	0.5437	0.9539	0.7903	0.8603	727.1480
Sep 21	0.5477	0.9463	0.7923	0.8568	727.6533
Oct 21	0.5453	0.9378	0.7933	0.8627	727.3577
Nov 21	0.5448	0.9297	0.7860	0.8512	727.8470
Dec 21	0.5449	0.9296	0.7870	0.8560	727.5513

Table 14. Annual average optical efficiency and output power — integrated result, Question 3.

Annual avg optical eff.	Annual avg cosine eff.	Annual avg shadow-blocking eff.	Annual avg truncation eff.	Annual avg output (MW)	Per-unit-area annual avg output (kW/m²)
0.5453	0.9443	0.7839	0.5878	61	727.455

Translator's note: *In Table 14 the "annual avg truncation efficiency" reads 0.5878, which is inconsistent with the ≈ 0.86 truncation values throughout the paper and looks like a transcription error. Reproduced as printed.*

Table 15. Design-parameter table — integrated result, Question 3.

Absorption-tower position	Heliostat size (W × H)	Install height (m)	Total number	Total mirror area (m^2)
(0, 0)	see Model 3	see Model 3	1745	see Model 3

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